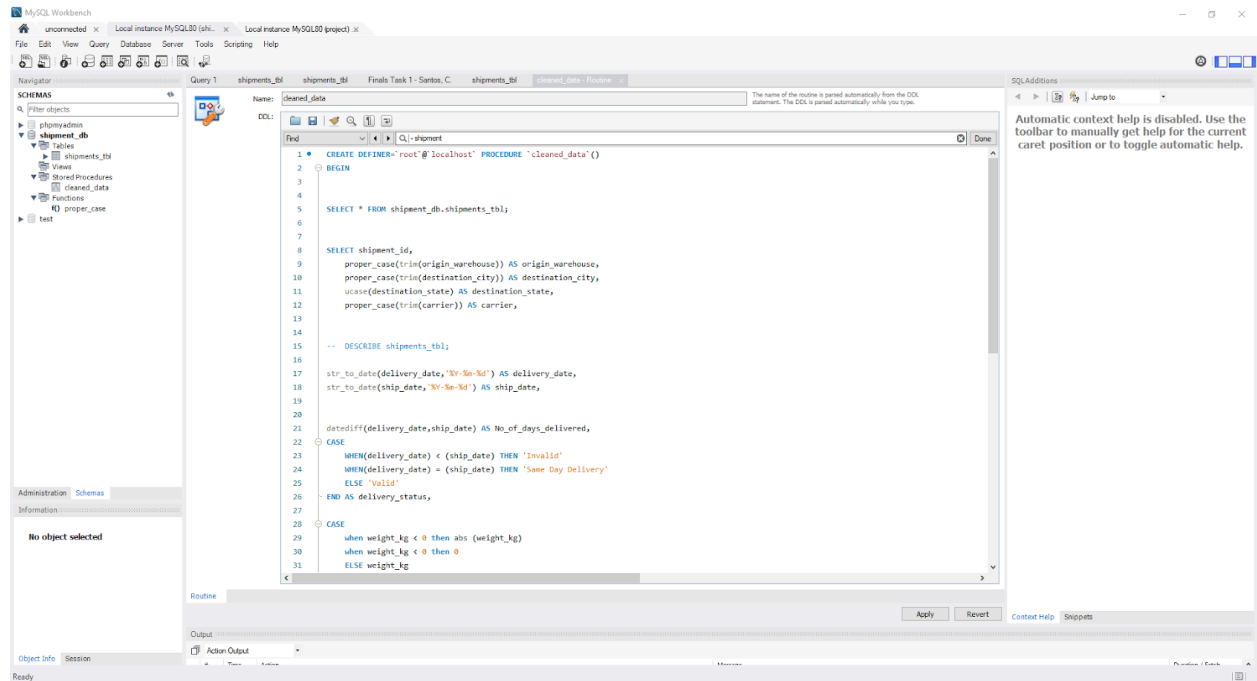


Finals PRACTICE Task 01

For this activity the data cleaning pipeline utilizes a MySQL stored procedure to standardize text fields and date formats, identify and delete duplicate records using window functions, and produce a clean, structured shipments dataset successfully validated in the workbench output.

Code 1



The screenshot displays the MySQL Workbench interface. The left sidebar shows the 'SCHEMAS' panel with a tree view containing 'shipment_db', 'shipments_tbl', 'views', 'stored_procedures', 'cleaned_data', 'functions', and 'test'. The main editor window is titled 'Query 1' and shows the SQL code for a stored procedure named 'cleaned_data'. The code is as follows:

```
1 CREATE DEFINER='root'@'localhost' PROCEDURE `cleaned_data`()
2 BEGIN
3
4
5 SELECT * FROM shipment_db.shipments_tbl;
6
7
8 SELECT shipment_id,
9        proper_case(trim(origin_warehouse)) AS origin_warehouse,
10       proper_case(trim(destination_city)) AS destination_city,
11       ucase(destination_state) AS destination_state,
12       proper_case(trim(carrier)) AS carrier,
13
14
15 -- DESCRIBE shipments_tbl;
16
17 str_to_date(delivery_date, '%Y-%m-%d') AS delivery_date,
18 str_to_date(ship_date, '%Y-%m-%d') AS ship_date,
19
20
21 datediff(delivery_date, ship_date) AS No_of_days_delivered,
22 CASE
23   WHEN (delivery_date < (ship_date)) THEN 'Invalid'
24   WHEN (delivery_date) = (ship_date) THEN 'Same Day Delivery'
25   ELSE 'Valid'
26 END AS delivery_status,
27
28 CASE
29   when weight_kg < 0 then abs(weight_kg)
30   when weight_kg < 0 then 0
31   ELSE weight_kg
```

Code 2

The screenshot shows the MySQL Workbench interface with a SQL routine named 'cleaned_data - Routine' open in the editor. The routine is a stored procedure that takes 'shipment_id' as an input parameter. It starts with a 'BEGIN' statement, followed by a 'CASE' statement that checks if 'weight_kg' is less than 0. If it is, it sets 'valid_weight' to the absolute value of 'weight_kg'; otherwise, it sets 'valid_weight' to 'weight_kg'. Then, it uses 'row_number()' over a partitioned set of rows from 'shipments_tbl' to assign 'row_num'. The partitioning is based on 'origin_warehouse', 'destination_city', 'destination_state', 'ship_date', 'carrier', 'weight_kg', and 'freight_cost'. The rows are ordered by 'shipment_id'. The routine then deletes rows from 'shipments_tbl' where 'shipment_id' is in the set of rows where 'row_num' is greater than 1. Finally, it ends with an 'END' statement. The interface includes a Navigator pane on the left showing the database schema, a central editor pane for the SQL code, and a right-hand pane for SQL additions and help.

```
DELIMITER //

CREATE PROCEDURE cleaned_data (shipment_id INT)
BEGIN
    CASE
        WHEN weight_kg < 0 THEN abs (weight_kg)
        WHEN weight_kg < 0 THEN 0
        ELSE weight_kg
    END AS valid_weight,
    row_number()
    over(
        partition by origin_warehouse, destination_city, destination_state,
        ship_date, carrier,
        CONVERT(weight_kg, char),
        CONVERT(freight_cost, char)
        ORDER BY shipment_id) as row_num
    FROM shipment_db.shipments_tbl)
    DELETE FROM shipments_tbl
    WHERE shipment_id IN
    (
        SELECT shipment_id FROM(
            SELECT shipment_id, ROW_NUMBER() OVER
            (PARTITION BY origin_warehouse,destination_city, carrier,
            ship_date ORDER BY shipment_id) as row_num
            FROM shipments_tbl
        ) as sub
        WHERE row_num > 1
    )
END
//
```

Output

The screenshot shows the MySQL Workbench interface with the output of the 'cleaned_data' routine displayed in the 'Result Grid' pane. The output is a table with columns: 'shipment_id', 'origin_warehouse', 'destination_city', 'destination_state', 'carrier', 'delivery_date', 'ship_date', 'No. of days delivered', 'delivery status', 'valid_weight', and 'row_num'. The table contains 25 rows of data, showing various shipments and their details. The interface also includes a Navigator pane on the left, a central editor pane, and a right-hand pane for SQL additions and help.

shipment_id	origin_warehouse	destination_city	destination_state	carrier	delivery_date	ship_date	No. of days delivered	delivery status	valid_weight	row_num
SHP-1001	Warehouse A	Dallas	TX	Feastflight	2024-01-14	2024-01-10	4	Valid	120.5	1
SHP-1002	Warehouse B	Los Angeles	CA	Speedyhaul	2024-01-11	2024-01-12	-1	Invalid	85.3	1
SHP-1004	Warehouse C	Chicago	IL	Feastflight	2024-01-20	2024-01-20	0	Valid	200	1
SHP-1005	Warehouse A	Houston	TX	Quickship	2024-01-20	2024-01-20	0	Valid	46.2	1
SHP-1006	Warehouse B	Phoenix	AZ	Speedyhaul	2024-01-23	2024-01-18	5	Valid	95	1
SHP-1007	Warehouse D	Seattle	WA	Quickship	2024-01-25	2024-01-20	5	Valid	310.8	1
SHP-1008	Warehouse C	Denver	CO	Feastflight	2024-01-25	2024-01-25	0	Valid	150	1
SHP-1009	Warehouse B	Miami	FL	Speedyhaul	2024-02-01	2024-01-25	7	Valid	88.7	1
SHP-1010	Warehouse A	Dallas	TX	Feastflight	2024-02-05	2024-02-01	4	Valid	175.3	1
SHP-1011	Warehouse D	Portland	OR	Quickship	2024-02-08	2024-02-03	5	Valid	60	1
SHP-1012	Warehouse B	Los Angeles	CA	Speedyhaul	2024-02-10	2024-02-05	5	Valid	85.3	1
SHP-1013	Warehouse C	IL	IL	Feastflight	2024-02-12	2024-02-08	4	Valid	225	1
SHP-1014	Warehouse D	San Francisco	CA	Quickship	2024-02-12	2024-02-12	0	Valid	140.6	1
SHP-1015	Warehouse A	Houston	TX	Feastflight	2024-02-16	2024-02-12	4	Valid	98.4	1
SHP-1016	Warehouse B	Phoenix	AZ	Speedyhaul	2024-02-19	2024-02-14	5	Valid	95	1
SHP-1017	Warehouse C	Denver	CO	Quickship	2024-02-23	2024-02-18	5	Valid	0	1
SHP-1018	Warehouse D	Seattle	WA	Quickship	2024-02-25	2024-02-20	5	Valid	310.8	1
SHP-1019	Warehouse A	Atlanta	GA	Feastflight	2024-02-27	2024-02-22	5	Invalid	165	1
SHP-1020	Warehouse B	Miami	FL	Speedyhaul	2024-03-02	2024-02-25	6	Valid	72.1	1
SHP-1021	Warehouse C	Chicago	IL	Feastflight	2024-03-05	2024-03-01	4	Valid	190	1
SHP-1022	Warehouse D	Portland	OR	Quickship	2024-03-08	2024-03-03	5	Valid	55.4	1
SHP-1023	Warehouse A	Dallas	TX	Feastflight	2024-03-09	2024-03-05	4	Valid	120.5	1
SHP-1024	Warehouse B	Los Angeles	CA	Speedyhaul	2024-03-10	2024-03-10	0	Invalid	130.2	1
SHP-1025	Warehouse C	San Francisco	CA	Quickship	2024-03-15	2024-03-10	5	Valid	146.8	1